



Differences in the innovation effectiveness of China's new energy vehicle industry policies: A comparison of subsidized and non-subsidized policies

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ABSTRACT

The adjustment of China's new energy vehicle (NEV) industry policies and innovation incentives is currently in progress. This study takes a new perspective by comparing subsidized and non-subsidized policies, and empirically examines their innovation effects on NEV enterprises using the innovation value chain theory. The results indicate that in the previous technology development stage, the subsidized policy exhibits a relatively stronger but volatile impact on incremental innovation, whereas the non-subsidized policy demonstrates a more stable effect on radical innovation. In the latter innovation value realization stage, both policies show a better impact on the short-term innovation value (output value) of NEV enterprises compared to the long-term value (market value). The mediation effect and threshold effect tests show that there is a significant mediating effect of technology development behavior of NEV enterprises in the whole value chain, although the specific mediating pathways and threshold effects are inconsistent, and overall innovation transmission efficiency is insufficient. Moreover, considering the internal ownership of NEV enterprises and external regional innovation atmosphere characteristics, the effects of the two policies also have their own distinct emphases. This study provides new insights for addressing issues related to innovation incentives and policy decisions for NEVs in the post-subsidy era.

1. Introduction

New energy vehicles (NEVs)¹ represent the primary direction for the transformation and green development of the global automobile industry. In recent years, China has achieved remarkable progress in the NEV sector, with both production and sales surpassed 9 million units in 2023. However, beneath the surface of this thriving market, China's NEV industry has been grappling with the dilemma of inadequate core technological capabilities [1,2]. Simultaneously, the subsidized policies initially implemented by the Chinese government, which primarily relied on direct financial subsidies, are facing significant challenges [3]. For some enterprises, the impetus to produce NEVs may not stem from genuine technological innovation or a commitment to green transformation, but rather from the pursuit of government subsidy funds to enhance their financial performance [4]. These distorted innovation outcomes, combined with the imperative for high-quality industrial development, prompted the Chinese government to propose a comprehensive tightening of NEV subsidized policies in 2016, marking the

onset of the post-subsidy era [5]. Due to the lack of sufficient market scale and innovation drive, the post-subsidy era does not mean that the government will completely withdraw its support. In this context, the government has shifted its focus toward the pivotal supplementary and potentially dominant role of non-subsidized policies for NEVs. These policies include market access catalogs, the dual-credit policy, and government procurement, which aim to heighten market access barriers, strengthen industry competition, and propel enterprise technological innovation [6]. Against this backdrop, a comprehensive analysis of the impacts and distinctions between subsidized and non-subsidized policies on the innovative development of NEV enterprises holds significant importance for addressing the innovation incentive challenges in the post-subsidy era and effectively leveraging the roles of these two policies.

The impact of NEV industry policies on technological innovation, particularly the role of subsidized policies, has been a contentious topic at the policy implementation level, sparking vigorous academic debates [7]. However, the discussion surrounding non-subsidized policies and

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¹ In the scope of China's NEV policy support, it mainly includes battery electric vehicles and plug-in hybrid electric vehicles.

their effects is still in the exploratory stage [8]. Furthermore, subsidized and non-subsidized policies often coexist in the NEV industry, yet their policy functions and tools are not consistent. Subsidized policies primarily involve government subsidies, tax reductions, and financial support, utilizing direct monetary compensation to address the lack of enterprise innovation investment and mitigate innovation risks [9,10]. In contrast, non-subsidized policies are characterized by their non-monetary nature and focus on supporting sustainable innovation development through infrastructure protection, market competition, demand support, and technology regulation [11]. This dichotomy between subsidized and non-subsidized policies can better reflect the actual development of China's NEV industry and the stage characteristics of its policy implementation. However, there is a dearth of research that clearly distinguishes and discusses the actual effects and differences between the two policies in promoting technological innovation in NEV enterprises, particularly in different stages of industrial development.

With respect to enterprise innovation outcomes, previous studies have predominantly analyzed the overarching or stage-based relationships between NEV policies and enterprise innovation input, behavior, or performance [12–14]. However, less emphasis has been placed on the entire value chain both before and after the innovation activities of NEV enterprises. From the viewpoint of technology innovation management research, the innovation value chain serves as a potent instrument for dissecting innovation activities and processes [15]. This framework postulates that enterprise innovation encompasses a comprehensive value chain, starting from innovation resource input, then progressing to the condensation of innovation knowledge, and ultimately culminating in the transformation of innovation outcomes [16]. Hence, to distinctly elucidate the efficacy and disparities of subsidized and non-subsidized policies in fostering innovation in NEV enterprises, it is necessary to consider the complete spectrum of enterprise innovation activities, i.e., examining the enterprise innovation value chain.

Based on the aforementioned research concepts, this study chooses listed NEV manufacturing enterprises in China and utilizes data from 2012 to 2022 as the research sample. Initially, a clear differentiation and selection of policy tools were made between NEV subsidized and non-subsidized policies. Drawing on innovation value chain theory, we developed a suitable research framework for the innovation process of NEV enterprises, integrating the aforementioned two policies for empirical testing. Additionally, we have taken into account the dynamic changes and stage-specific characteristics of the two policies, as well as the potential disparate innovation outcomes that may arise from internal or external factors of NEV enterprises during the analytical process.

This study makes several contributions. 1) This study focuses on the effectiveness and differences of subsidized and non-subsidized policies on the technological innovation of NEV enterprises. While there is a growing body of literature on government subsidies and their impact on innovation in the NEV industry, relatively few studies have comprehensively discussed the differences in the effects of these two types of policies from the perspective of non-subsidized policies. The research theme can provide a valuable supplement to the current analytical framework. 2) Utilizing innovation value chain theory, this study explores the technological innovation process and pathways of NEV enterprises under the influence of subsidized and non-subsidized policies. The innovation value chain framework and its evaluation indicators can clearly demonstrate the entire process from policy response to technology development and then to the innovation output and value creation of NEV enterprises. It also highlights the differences in the effectiveness of the two policies in different pathways. This provides empirical evidence for accurately leveraging the effective role of NEV policies in the post-subsidy era. 3) This study found that, compared to the subsidized policy, the non-subsidized policy has a greater and more stable incentive effect on the radical innovation behavior of NEV enterprises. Additionally, the policy's stimulation of radical innovation leads to a more pronounced effect in promoting the long-term value (market value) realization of enterprise innovation. This provides

valuable insights for further solidifying the direction of NEV policy development and optimizing the innovation incentive system for NEV enterprises.

2. Literature review and theoretical framework

2.1. Literature review

(1) Two types of policies and effects. The existing theoretical research on NEV subsidized policies is relatively mature and has focused mainly on the incentive effects of government direct subsidies and tax incentives [10,17]. Scholars have acknowledged the positive effects of subsidized policy innovation incentives, such as stimulating relevant enterprises to increase R&D investment, effectively promoting enterprise financing, alleviating the uncertainty of high investment in R&D activities, and further expanding the innovation demand market for NEVs [18–20]. However, some scholars argue that NEV subsidized policies also have inevitable flaws, such as a lack of pertinence and sustainability, difficulty in policy implementation and supervision, and the potential to cause rent-seeking behavior and vicious competition [21–23]. Therefore, it is crucial to distinguish their incentive effects and clearly explain their true effectiveness for NEV enterprise innovation to avoid excessive subsidies and subsidy dependence.

Compared to the well-established research system on subsidized policies, research on non-subsidized policies for NEVs is still in its early stages. Some scholars have proposed concepts such as non-financial incentives [24], non-innovation subsidies [14], non-subsidized policies [11], and non-monetary policies [25], which have expanded the scope of NEV policies beyond government subsidies. However, there is still a lack of a unified theoretical basis and academic terminology in this area. Regarding the specific roles of NEV non-subsidized policies, studies have revealed that these policies can encompass a wider range of tools and focal points, including market diffusion and consumer adoption, enterprise production and performance, technological innovation process and management, environmental protection and sustainable development. This broad scope covers the macro-industrial policy and shows a complementary relationship with NEV subsidized policies [26]. However, there are still several issues in the existing research on NEV non-subsidized policies. For instance, the current research is often scattered and primarily focuses on the effects of single policies or treats non-subsidized policies as auxiliary measures to subsidized policies [27]. These include such as the dual-credit policy [28,29], infrastructure construction policy [30,31], commercial demonstration and promotion policy [32,33], government procurement policy [25,34], and right-of-way priority policy [35,36]. Moreover, there is a lack of systematic empirical comparisons between subsidized and non-subsidized policies, particularly in terms of their effects on innovation incentives.

(2) Two types of policies and tool choices. First, referring to mature practices, the subsidized policy primarily employs two commonly used policy tools: government subsidies and tax incentives. Government subsidies, as unpaid government expenditure items, are earmarked for specific funds. Enterprises that receive subsidies often classify these funds as R&D investments, reserving them for technological innovation. Tax incentives, on the other hand, are benefits obtained after the enterprise's innovation results have been tested by the market. They can offset part of the R&D costs and alleviate the enterprise's innovation risk. These two policy tools can serve as ex-ante and ex-post incentives, respectively [10].

Second, there is no standardized approach for selecting NEV non-subsidized policy tools. Referring to the research and statistics [6], there are three main types of non-subsidized policy tools that have been officially released in China, which are closely related to incentivizing

enterprise innovation: NEV model access and application catalogs (the market access policy²), the dual-credit policy³ (*Passenger Vehicle Corporate Average Fuel Consumption Credits and New Energy Vehicle Credits*), and government or public preference procurement (the government procurement policy⁴). Also from the perspective of ex-ante and ex-post incentives, the market access policy relies mainly on product technical specification reviews and access systems to increase the market access threshold. In terms of ex-ante incentives, it leverages policy screening and elimination characteristics to stimulate the output of qualified innovation products from NEV enterprises. The dual-credit policy and government procurement measures are more focused on the final products of NEV enterprises. The dual-credit policy mandates the proportion of NEV production and sales for each vehicle enterprise, which is used to carry out credit quotas and trading. This offsets enterprise innovation costs through ex-post market competition and credit trading revenue. The government procurement policy is more ex-post related, with established standards and demand directly promoting the growth of NEV consumption, and driving targeted innovation and innovation diffusion by enterprises. Therefore, the above is the basic division of NEV subsidized and non-subsidized policies and their policy tool choices.

In summary, existing research has indicated controversy surrounding the innovation role of subsidized policies for NEVs, and attention has begun to focus on non-subsidized policies and their effects. However, there is still room for further expansion. On the one hand, previous studies, whether from the supply side or demand side of industrial policy, or selective and functional perspectives, have focused on the role of subsidized policies or analyzed them primarily based on government subsidies. Studies on the role of non-subsidized policies and comparisons of these two policies at the level of technological innovation incentives for NEV enterprises are rare. On the other hand, enterprise technological

innovation activities under policy influence form a closely connected systemic process. From a theoretical perspective, the transition of emerging technology to large-scale industrialization requires significant efforts from enterprises within the industry. In practice, the preliminary results of NEV enterprise innovation response and technology development must be transformed as much as possible into high output performance and high market value, forming an effective closed loop from policy resource input to innovation technology development, and then to the realization of innovative value, which is what constitutes meaningful innovation. This is also the ultimate goal of the transformation of subsidized policies and the relay of non-subsidized policies in the future, that is, stimulating the coordinated realization of short-term innovation growth and long-term innovation value in NEV enterprises. Therefore, this study focused on a detailed analysis of the aforementioned shortcomings.

2.2. Theoretical framework and research hypothesis

Innovation value chain theory proposes that enterprise innovation activities can be conceptualized as a complete value chain, encompassing the acquisition and input of innovation resources, intermediate and final outputs, and ultimately value realization [16]. It follows the basic logic of “innovation drive - innovation output - innovation value” [37]. Building upon this theoretical foundation, Chen et al. [15] simplified the innovation value chain in the high-tech industry into two subprocesses: the R&D process and the commercialization process. Yuan and Li [38] further incorporated government policies as a pivotal link, highlighting that public policies (market pull and technology push) can facilitate the formation and evolution of innovation value chains in industries, albeit with a certain time lag. In the context of NEVs, He et al. [29] used the dual-credit policy as an illustrative example to explore the impact of NEV policies on manufacturers' green technology innovation value chain, which also encompasses the fundamental processes of R&D and innovation transformation. Therefore, we apply the innovation value chain theory to our research topic and construct an evaluation framework. This framework is divided into two key stages, the R&D process and the commercialization process, as illustrated in Fig. 1.

- (i) The previous stage: policy resource input - enterprise technology development. As China's NEVs do not yet have an independent competitive advantage in the market, they still rely on government policies and related resources for support. Various types of industrial policies, including subsidized and non-subsidized policies, constitute the primary link to innovation resources invested by NEV enterprises. In response to these policies, NEV enterprises will initially adopt different policy responses and innovation behavior strategies [11]. They will then convert the invested innovation resources into knowledge accumulation, leading to primary technology development outcomes such as patent applications and the establishment of new technology production lines. This will provide feedback on the initial impact of industrial policies from the technology mid-range and complete the R&D process of the previous stage. Furthermore, the tool composition and operational principles of subsidized and non-subsidized policies for NEVs are not consistent, and this disparity will also manifest in the specific technology development of NEV enterprises. Based on the above analysis, the following hypothesis can be proposed.

H1. Both subsidized and non-subsidized policies have a promoting effect on the technology development of NEV enterprises in the previous stage, and there will be differences in the specific effects of the two policies.

² **The market access policy.** As a representative of ex-ante intervention policies in technological innovation, the market access for NEVs began with the issuance of the first batch of the *Recommended Model Catalog for Energy-saving and NEV Demonstration and Application Project by the Ministry of Industry and Information Technology of China* in August 2009. Subsequently, on a monthly basis, a new batch of NEV manufacturers and their vehicle models that meet the market access conditions are released to the public, along with detailed information on the technical standards and composition of the vehicles. This has led to the formation of a government-led promotion catalog for NEVs. NEV products that are not included in the promotion catalog are not eligible for relevant national financial and tax support, making it difficult for them to enter the market.

³ **The dual-credit policy.** As a crucial component of ex-post guidance policies in technological innovation, the dual-credit system for NEVs comprises two types of credits: *Corporate Average Fuel Consumption Credit (CAFC)* and *New Energy Vehicle Credit (NEVC)*. Under this system, companies with positive CAFC can carry forward or transfer their credits to affiliated enterprises, while positive NEVC points can be freely traded. If a vehicle manufacturer fails to meet the mandatory percentage of NEV production and sales as stipulated in the dual-credit policy, resulting in low or negative NEVC, it must purchase positive NEVC from other manufacturers to compensate for its credit disadvantage and avoid policy penalties. This creates an indirect guidance mechanism that emphasizes the importance of achieving credit, market transactions, and innovative compensation strategies.

⁴ **The government procurement policy.** As another important component of ex-post guidance policies, the NEV scale procurement policy by the government and public institutions can be traced back to as early as 2009. Subsequently, the central government encouraged local governments to purchase NEVs for public transportation, sanitation, postal service, and public service through administrative control, special subsidies, and pilot city demonstrations. This helped to gradually expand the scope of government (public) procurement and establish an NEV procurement standard system. As a result, the priority adoption and demonstration effect was formed in the whole society, and the industry's innovation ability was improved through public market demand, thus achieving the goal of demand-induced innovation.

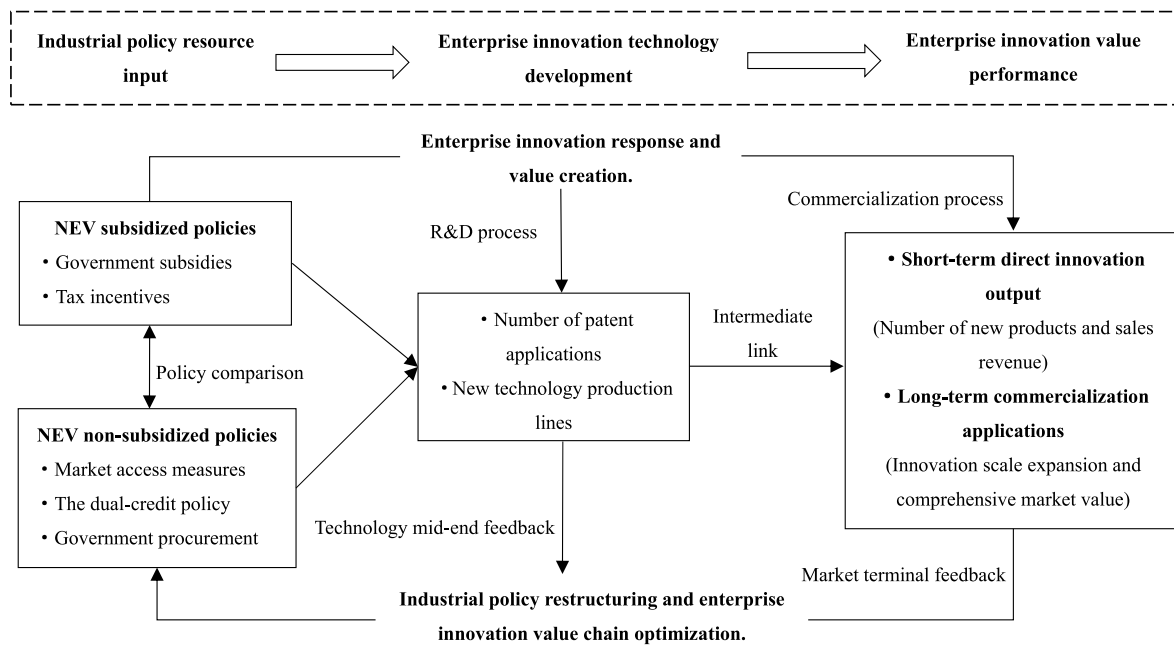


Fig. 1. The innovation value chain framework of NEV enterprises.

(ii) The latter stage: policy resource input - enterprise innovation value. Depending on different technology development strategies and initial output results, the performance of enterprise innovation value reflects the process by which technology development outcomes influenced by previous industrial policies, including subsidized and non-subsidized policies, are integrated into tangible NEV products and introduced to the market to generate profits. Meanwhile, the input and output results of innovation policy resources at various stages of the value chain do not progress synchronously. This process will showcase short-term direct innovation outputs, such as the introduction of new products and increased sales revenue, as well as long-term commercialization applications, such as the expansion of innovation scale and the overall market value of enterprises [1]. These aspects signify the ultimate realization of the enterprise's innovation value and commercialization journey from a market perspective. Furthermore, the distinct effects of the two policies at different value realization stages can also offer valuable insights for the refinement of industrial policies. Based on the above analysis, the following hypothesis can be proposed.

H2. Both subsidized and non-subsidized policies have a promoting effect on the innovation value of NEV enterprises in the latter stage, and there will be differences in the specific effects of the two policies.

(iii) The intermediate link: enterprise technology development. Based on the theoretical framework constructed, enterprise technology development is the pivotal variable of the entire innovation activities of NEV enterprises. It represents both the output of innovation from the initial R&D process and the input for innovation in the subsequent commercialization phase [29]. Both subsidized and non-subsidized policies lead to initial decisions for enterprise innovation responses, and these policies can impact the final output and value level of innovation by influencing the way and extent of enterprise technology development. This, in turn, facilitates the effective transference of policy effects throughout the entire innovation value chain, forming a closed

loop. Thus, the technology development activities of NEV enterprises assume a central intermediary role between industrial policies and innovation value. Furthermore, due to the temporal disparities in realizing short-term and long-term innovation value, enterprise technology development in the intermediate link may also exhibit nonlinear threshold effects [39]. Consequently, the following hypotheses can be proposed.

H3. There is a mediating effect of enterprise technology development between NEV policies (subsidized and non-subsidized policies) and innovation value.

H4. There is a threshold effect of enterprise technology development between NEV policies (subsidized and non-subsidized policies) and innovation value.

3. Research design

3.1. Research sample and variable selection

In 2012, the production and sales of NEVs in China surpassed 10,000 units, marking the beginning of a period of rapid growth and commercialization. Considering these two policies, especially the implementation of non-subsidized policies, the main target is still domestic vehicle manufacturing enterprises. Therefore, we selected data from listed NEV manufacturers in China from 2012 to 2022 as the research sample. The sample consists of 14 state-owned enterprises and 10 non-state-owned enterprises. We obtained enterprise data primarily from the CSMAR database, the China State Intellectual Property Office, and the annual reports of each enterprise. Industrial policy data were sourced from the China Energy Conservation and New Energy Vehicle Yearbook, the official website of the Ministry of Industry and Information Technology, and the China Government Procurement Network.

Dependent variables: enterprise innovation value (short-term value and long-term value). Previous studies have primarily measured enterprise innovation value using dimensions such as innovation output, performance, or efficiency. To account for the time lag between enterprise innovation output and value realization, and overcome the

limitations of single indicators in measuring the effect of industrial policy on enterprise innovation value, this study employs a two-phased approach. This includes measuring short-term direct value (SValue) and long-term indirect value (LValue) to assess the innovation value of NEV enterprises [1,29]. The short-term value is typically reflected in direct indicators of innovation output, such as the number of innovative products or the sales revenue of an enterprise. Due to data availability, we utilize the number of new product outputs (the output value) of each NEV enterprise published by the *Ministry of Industry and Information Technology of China* as a measure of SValue. The long-term value is generally reflected in indirect comprehensive indicators, such as the expansion of innovation scale and market value performance. Considering the data availability, we use the market value of the enterprise as the measurement index for long-term value [40]. Specifically, we express this using the Tobin Q value (A value) of each listed enterprise in the CSMAR database. In practice, we treat the variable SValue with a one-year lag and the variable LValue with a two-year lag [39].

Independent variables: the subsidized policy and the non-subsidized policy. There are several common methods for measuring industrial policy. One approach is to measure industrial policies by the specific values of government subsidies, tax incentives, and financial support. Another method is to count the number of administrative regulations and departmental rules related to the implementation of industrial policies and rank scoring them accordingly. A third method is to use dummy variables to study the impact of industrial policies based on whether the introduction of government policies encourages the development of key industries or enterprises. In this study, the NEV subsidized policy (Pol_{sub}) mainly consists of two policy tools: government subsidies and tax incentives. Therefore, referring to the first method, the measurement index is the cumulative value of government innovation subsidies and tax rebates received by the enterprise in that year, which is subsequently standardized. The NEV non-subsidized policy ($Pol_{non-sub}$) mainly includes market access, the dual-credit, and government procurement measures. Due to the specificity of these policy tools and their different assessment objectives, it is challenging to measure the overall effect of the non-subsidized policy using any of these individual methods. Moreover, the non-subsidized policy is more selective, and some NEV enterprises may not benefit from certain policy tools in some years. To avoid the endogeneity problem, this study combines the second and third methods based on existing specific practices [6]. Specifically, dummy variables are used to indicate whether an enterprise obtains a market access policy, dual-credit policy, or government procurement policy in a given year. The total score of the three policy tools obtained by the enterprise in that year is then calculated, which serves as the measurement index of the non-subsidized policy ($Pol_{non-sub}$). It is also standardized in practice.

Mediating variable: enterprise technology development (TD). In measuring TD, previous research has suggested that data on enterprise patent applications can serve as a suitable proxy indicator, representing the initial technological achievements in the R&D stage of the innovation value chain [29]. Patent applications can ensure the originality of innovation and increase the likelihood of realizing market value [41]. Therefore, we utilize the number of patent applications filed by NEV enterprises as a measure of TD. Furthermore, enterprise patent applications can be classified into two types, radical innovation and incremental innovation, allowing us to explore which type of innovation has a more significant mediating effect. Consistent with established practices [32,42], we define the number of invention patent applications filed by the enterprise in a year as radical innovation (TD1), while the number of non-invention patents (utility model and design patents) applications represents incremental innovation (TD2).

Control variables. Considering the potential impact of factors such as enterprise characteristics and financial performance on innovation outcomes, and following established research practices [11,43], several key indicators have been identified as control variables. These include enterprise size (Size), age (Age), ownership structure (Own), return on

assets (ROA), asset-liability ratio (ALR), and human capital (HC). The Size indicator represents the total asset value of the enterprise (hundred million yuan) and has been transformed using the natural logarithm. The Age indicator is defined as the number of years since the establishment of the enterprise, also transformed using the natural logarithm. The Own indicator is captured through a dummy variable, assigning a value of 1 for state-owned enterprises and 0 for non-state-owned enterprises. The ROA indicator is calculated as the net profit of the enterprise divided by the total asset balance. The ALR indicator is calculated as the total liabilities of the enterprise divided by the total asset balance. The HC indicator is calculated as the number of technical personnel in the enterprise divided by the total number of employees.

3.2. Econometric models

To analyze the overall impact of subsidized and non-subsidized policies on the innovation value chain of NEV enterprises, we constructed a basic fixed-effect model [20,44]. Model (1) represents the overall impact of the two policies on enterprise technology development in the previous stage, and models (2) and (3) represent the overall impact on enterprise innovation value in the latter stage.

$$TD1_{i,t}(TD2_{i,t}) = \lambda_0 + \lambda_1 Pol_{sub,i,t} + \lambda_2 Pol_{non-sub,i,t} + \sum Controls_{i,t} + \varepsilon_{i,t} + \mu_i, \quad (1)$$

$$SValue_{i,t} = \beta_0 + \beta_1 Pol_{sub,i,t-1} + \beta_2 Pol_{non-sub,i,t-1} + \sum Controls_{i,t-1} + \varepsilon_{i,t} + \mu_i, \quad (2)$$

$$LValue_{i,t} = \beta_0 + \beta_1 Pol_{sub,i,t-2} + \beta_2 Pol_{non-sub,i,t-2} + \sum Controls_{i,t-2} + \varepsilon_{i,t} + \mu_i. \quad (3)$$

Next, to validate the rationality of the innovation value chain process in NEV enterprises, we incorporate TD1 and TD2 as mediating variables to examine the mediating role of enterprise technology development under the influence of two policies. Building upon models (1)–(3), we introduce the mediating variables to construct models (4) and (5), and the Bootstrap method was used to test the existence of mediating effects in steps [43].

$$SValue_{i,t} = \beta'_0 + \beta'_1 Pol_{sub,i,t-1} + \beta'_2 Pol_{non-sub,i,t-1} + \varphi_1 TD1_{i,t-1}(TD2_{i,t-1}) + \sum Controls_{i,t-1} + \varepsilon_{i,t} + \mu_i, \quad (4)$$

$$LValue_{i,t} = \beta'_0 + \beta'_1 Pol_{sub,i,t-2} + \beta'_2 Pol_{non-sub,i,t-2} + \varphi_2 TD1_{i,t-2}(TD2_{i,t-2}) + \sum Controls_{i,t-2} + \varepsilon_{i,t} + \mu_i. \quad (5)$$

In the above model, i is the firm, t is the year, $\varepsilon_{i,t}$ is the random disturbance term, μ_i represents the fixed firm. The descriptive statistics and correlation coefficients for each variable can be found in Appendix A. For the main models (1)–(3), the Hausman test was conducted to ensure their validity. When the enterprise technology development indicators (TD1, TD2) and innovation value indicators (SValue, LValue) were considered as dependent variables, the Hausman test results yielded p -values of 0.0002, 0.0005, 0.0097, and 0.0056, respectively, all of which were significant at the 1 % level, indicating that the fixed-effect models utilized in this study are appropriate. Moreover, tests for multicollinearity and serial correlation were conducted using Variance Inflation Factors (VIF) and Durbin-Watson (DW) statistics. The results indicated that in each model, the maximum individual VIF value was 2.72, with average VIF values of 1.47, 1.47, 1.63, and 1.56, all well below the threshold of 5. Additionally, the DW values were 1.893, 1.856, 1.773, and 1.817, all around 2, suggesting the absence of significant multicollinearity and autocorrelation concerns within the models.

4. Empirical results and discussion

4.1. Innovation incentive effects of subsidized and non-subsidized policies

4.1.1. Overall impact

Based on models (1)–(3), we can initially derive the overall impact results of two policies on technology development (TD1, TD2) and innovation value (SValue, LValue) for NEV enterprises, as presented in Table 1.

(1) Impact on the previous stage of enterprise technology development. In terms of overall technology development (TD), Table 1 (1a) indicates that the Pol_{sub} and $Pol_{non-sub}$ coefficients are 0.259 and 0.174, respectively. They are significantly positive at the 5 % or 10 % level, suggesting that both subsidized and non-subsidized policies can effectively promote the overall technology development of NEV enterprises, as represented by patent applications.

Furthermore, when the overall TD indicator is replaced with specific indicators for radical innovation (TD1) and incremental innovation (TD2), the results are presented in Table 1 (1b) and (1c). For TD1, part (1b) shows that the Pol_{sub} coefficient is positive but not significant (0.116), while the $Pol_{non-sub}$ coefficient is positive and significant at the 5 % level (0.209). This finding suggests that the existing non-subsidized policy has a relatively stronger incentive for high-level radical innovations in the technology development stage. However, for TD2, the policy effect is the opposite. Part (1c) shows that Pol_{sub} is positively significant at the 5 % level with a relatively large coefficient (0.278), and $Pol_{non-sub}$ is positively significant at the 10 % level with a relatively small coefficient (0.141). This indicates that compared to the non-subsidized policy, the existing subsidized policy is more likely to stimulate more conservative and practical incremental innovation in NEV enterprises, and the policy effect is more pronounced. Therefore, hypothesis H1 is confirmed.

(2) Impact on the latter stage of enterprise innovation value. Table 1 (2a) demonstrates that both policies (Pol_{sub} , $Pol_{non-sub}$) have positive significant coefficients at the 1 % or 10 % level in the short-term value (SValue), with the Pol_{sub} coefficient (0.315) being greater than the $Pol_{non-sub}$ coefficient (0.106). This finding suggests that the

two policies can significantly promote the short-term innovation value of NEV enterprises, as represented by the direct output of new products, with the subsidized policy having a greater effect. However, for the long-term value (LValue, 2b), the Pol_{sub} coefficient is negative (−0.040) and not significant, the $Pol_{non-sub}$ coefficient is positively significant (0.080) at the 5 % level but lower than the short-term value. This indicates that, compared to the short-term output value, the two policies have a lower promotion effect on the long-term market value of NEV enterprises, and even the subsidized policy shows a certain tendency to hinder. Therefore, hypothesis H2 is confirmed.

4.1.2. Dynamic impact

Building upon the overall impact, further analysis was conducted to examine the dynamic impact and differences of the subsidized and non-subsidized policies. Based on models (1)–(3), the individual data were accumulated and averaged within years. By utilizing time-varying parameter models and the Kalman filtering algorithm, continuous estimates of the state variables over time were obtained [45]. These represent the dynamic impact results of the key independent variables on the dependent variables, as illustrated in Figs. 2 and 3.

(1) Dynamic impact on the previous stage of enterprise technology development. For the overall technology development (TD), Fig. 2a illustrates that both policies' innovation incentives exhibit a phase fluctuation. It can be divided into three phases: the subsidized policy dominance period (2012–2016), the two policies' parallel period (2017–2019), and the non-subsidized policy leading period (2020–2022). Notably, the subsidized policy exhibits a more pronounced volatility in its impacts.

Firstly, during the subsidized policy dominance period (2012–2016), the effects of the two policies exhibited a continuous upward trend, reaching their peak in 2016, with the subsidized policy taking a dominant role. This can be attributed to the initial stage of industrial development, where the government implemented large-scale policy stimuli in the form of direct monetary relief. The expansion of financial subsidies and increasing consumer demand had a significant impact on demand-induced innovation in the NEV industry for a short period.

Next, during the two policies' parallel period (2017–2019), the innovation incentives of both policies decreased, following parallel trends. Several factors may contribute to this trend. Following the outbreak of large-scale subsidy fraud incidents by China's NEV enterprises in 2016, the government initiated a subsidy withdrawal plan. The sudden reduction in subsidy benchmarks exceeded market producers' expectations, leading to a more pronounced decrease in the innovation incentives provided by the subsidized policy. For the non-subsidized policies, such as model catalog access, the dual-credit trading and competition, have been significantly strengthened since 2017. Consequently, although the innovation effect of the non-subsidized policy also declined, the magnitude of the fluctuation was relatively smaller.

Finally, during the non-subsidized policy leading period (2020–2022), there was a fundamental shift in the innovation effects of the two policies, with the non-subsidized policy taking the lead. This shift can be attributed to the Chinese government's original plan to completely withdraw direct consumption subsidies for NEVs by the end of 2020 (later extended to the end of 2022). This withdrawal significantly enhanced the timeliness of non-subsidized policy relays. Additionally, the global outbreak of COVID-19 had a substantial impact on China's NEV industry and technological innovation. The retrenchment of government financing further dampened the innovative role of the subsidized policy.

Regarding specific radical innovation (TD1) and incremental innovation (TD2), as illustrated in Fig. 2b, both policies demonstrate similar three-stage patterns in their impact. Moreover, this can be further delineated into the subsidized policy dominance period for incremental

Table 1
The overall impact results.

Variables	Technology development			Innovation value	
	(1a) TD	(1b) TD1	(1c) TD2	(2a) SValue	(2b) LValue
Pol_{sub}	0.259** (0.129)	0.116 (0.185)	0.278** (0.133)	0.315*** (0.110)	−0.040 (0.060)
$Pol_{non-sub}$	0.174* (0.104)	0.209** (0.099)	0.141* (0.081)	0.106* (0.057)	0.080** (0.039)
Size	0.618*** (0.175)	0.609*** (0.170)	0.635*** (0.184)	0.327*** (0.224)	−0.204*** (0.057)
Age	−0.676* (0.484)	0.320 (0.397)	−1.393*** (0.427)	−0.755*** (0.224)	−0.001 (0.125)
Nature	−0.553 (0.489)	−0.803* (0.475)	−0.607 (0.512)	−0.121 (0.292)	−0.118 (0.174)
ROA	−0.501 (1.130)	0.044 (1.099)	−0.736 (1.184)	1.242 (0.890)	0.468 (0.508)
ALR	1.518* (0.853)	2.503*** (0.830)	1.233 (0.894)	0.214 (0.391)	−0.261 (0.201)
HC	0.971* (0.544)	0.561 (0.515)	0.902 (0.594)	0.727 (0.530)	0.381 (0.298)
Constant	3.437*** (1.239)	−0.453 (1.204)	4.704*** (1.297)	3.218*** (0.436)	2.890*** (0.341)
Firm FE	Yes				
N	260	260	260	260	260
Adj. R ²	0.799	0.834	0.750	0.860	0.348

Note: ***, **, and * represent significance at the 1 %, 5 %, and 10 % levels, (standard error), same below.

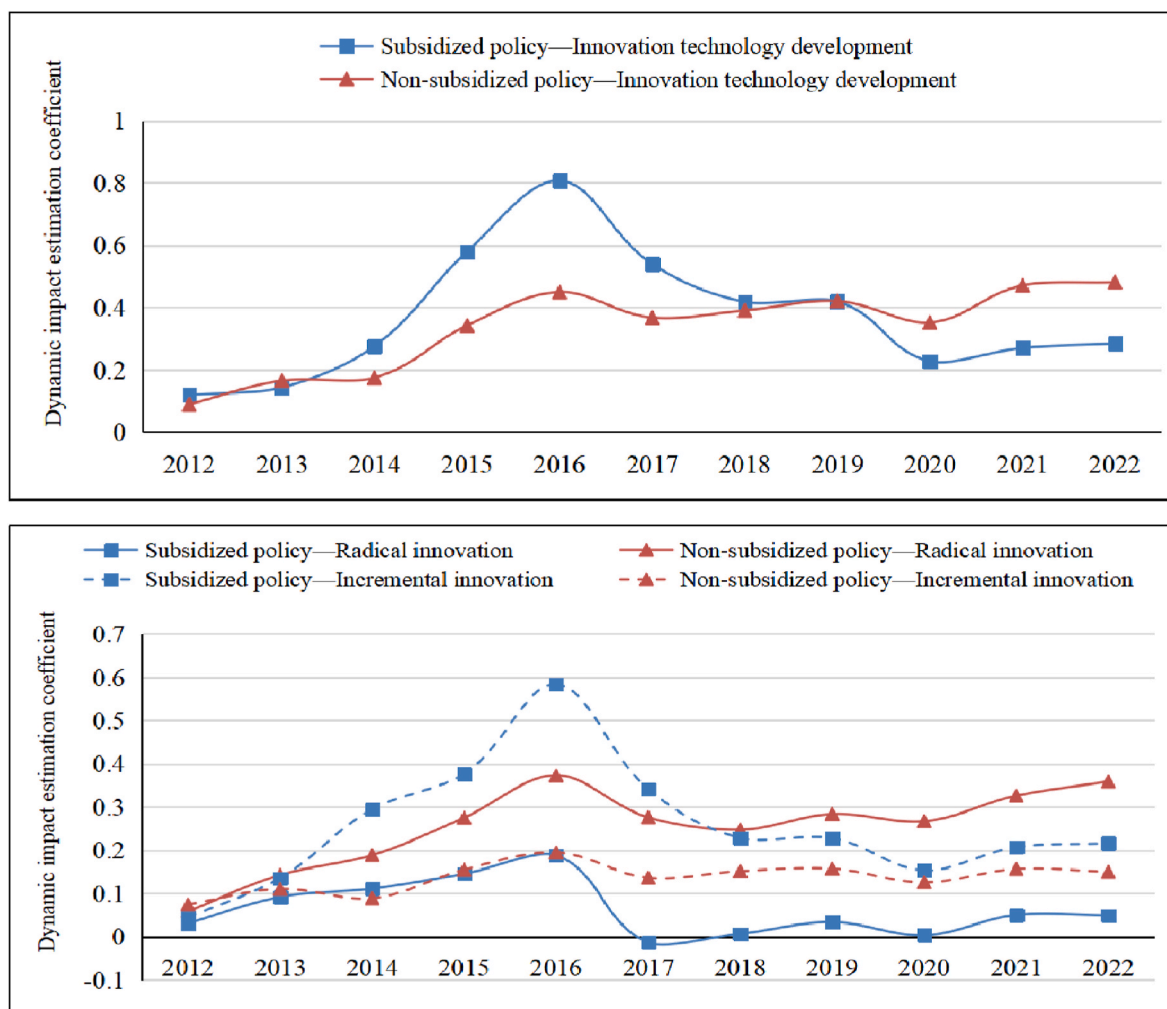


Fig. 2. The dynamic impact results of the previous stage.

Fig. 2a. Overall innovation technology development.

Fig. 2b. Specific radical innovation and incremental innovation. .

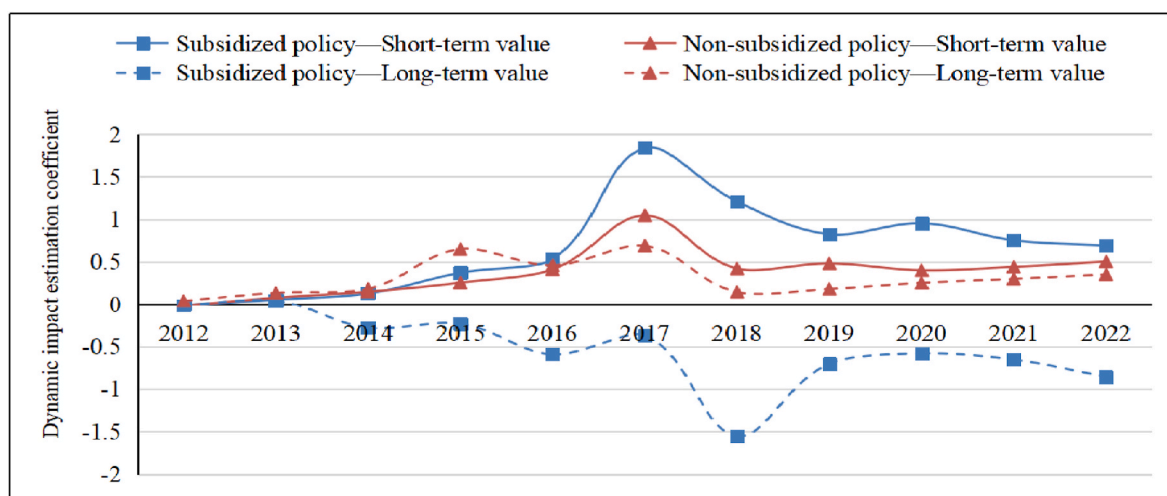


Fig. 3. The dynamic impact results of the latter stage.

innovation (2012–2016); the parallel period of the subsidized policy for incremental innovation and the non-subsidized policy for radical innovation (2017–2019); and the non-subsidized policy leading period for radical innovation (2020–2022).

(2) Dynamic impact on the latter stage of enterprise innovation value. Considering the lag in the realization of enterprise innovation value and the specific settings of the variables SValue lagging one year and LValue lagging two years, the results are presented in Fig. 3. Analyzing the results within the framework of the three-stage segmentation, it is evident that the differentiated impact of the subsidized policy is more pronounced, displaying a two-tier trend. Conversely, the impact of the non-subsidized policy appears to be more stable and consistent over time.

Specifically, the first stage is primarily characterized by the onset of the impacts of the two policies (2012–2016). Throughout this phase, the effects of the policies remain relatively modest, with distinctions not yet clearly delineated. It is worth noting that the subsidized policy exhibited a certain inhibitory effect on the long-term innovation value (market value) right from the outset.

The second stage is primarily characterized by an obvious differentiation in the impact of the subsidized policy (2017–2019). With the extensive utilization of large-scale financial tools in the early stages, the influence of the subsidized policy on the short-term innovation value (output value) of NEV enterprises began to escalate rapidly. This influence peaked in 2017 and played a pivotal role in shaping the overall policy impact. However, the impact of the subsidized policy also displayed a polarizing trend, as its effect on long-term value declined rather than increased, indicating a greater impediment. This underscores that while simple monetary subsidies can swiftly stimulate the direct product output of NEV enterprises in the short term, they encounter challenges in enhancing the long-term market value of enterprise innovation and may even weaken their overall competitive prowess. Concurrently, the non-subsidized policy also offers stronger incentives for short-term innovation value compared to long-term value during this period.

The third stage is primarily characterized by the stabilization of the impact of the non-subsidized policy (2020–2022). With the complete cessation of direct subsidies for NEVs and the progressive strengthening of non-subsidized policies, the polarized nature of the subsidized policy's impact persists, continuing to prioritize short-term output value. Concurrently, the innovation influence of the non-subsidized policy shows a tendency towards stability, with its effect on long-term value gradually approaching the short-term value. This trend signifies a good outlook for future development.

4.2. The mediating effect and threshold effect of enterprise technology development

(1) Mediating effect. According to models (4) and (5), the mediating effect test results for TD1 and TD2 are shown in Table 2. In the process of the subsidized policy's influence, both radical innovation (TD1) and incremental innovation (TD2) exhibit a significant partial mediating effect only on the Pol_{sub}-SValue path (3a, 5a), with the mediating effect proportion of TD2 (0.137) being significantly greater than that of TD1 (0.030).

In the process of the non-subsidized policy's influence, there are differences in TD1. TD1 is partially mediated by both paths of Pol_{non-sub}-SValue and Pol_{non-sub}-LValue (4a, 4b), with the mediating effect on long-term value (LValue) being relatively high (4b, 0.204). On the other hand, TD2 is partially mediated only in the Pol_{non-sub}-SValue path (6a). As a result, hypothesis H3 receives preliminary validation.

(2) Threshold effect. The above mediating effects (hypothesis H3) demonstrate that the various types of TD have inconsistent

Table 2
The mediating effect test results.

Mediating variable: TD1												
(3a)			(3b)			Variables		(4a)		(4b)		
Variables	SValue	TD1	SValue	LValue	TD1	LValue		SValue	TD1	SValue	TD1	LValue
Pol _{sub}	0.315*** (0.110)	0.116 (0.185)	0.301*** (0.110)	-0.040 (0.060)	0.116 (0.185)	-0.029 (0.061)	Pol _{non-sub}	0.106* (0.057)	0.209** (0.099)	0.079* (0.046)	0.080*** (0.025)	0.209** (0.099)
TD1			0.082* (0.046)			0.078*** (0.035)	TD1			0.082* (0.046)	0.078*** (0.035)	
Mediating type	Partial mediating effect						Mediating type	Partial mediating effect				
Mediating effect/ Total effect	0.030						Mediating effect/ Total effect	0.162			Partial mediating effect 0.204	
Mediating variable: TD2												
Variables	(5a) SValue	TD2	SValue	(5b) LValue	TD2	LValue	Variables	(6a) SValue	TD2	SValue	(6b) LValue	LValue
Pol _{sub}	0.315*** (0.110)	0.278** (0.133)	0.238** (0.105)	-0.040 (0.060)	0.278** (0.133)	-0.302 (0.111)	Pol _{non-sub}	0.106* (0.057)	0.141* (0.081)	0.096** (0.047)	0.080*** (0.025)	0.141* (0.081)
TD2			0.155** (0.079)			0.048 (0.042)	TD2			0.155** (0.079)		
Mediating type	Partial mediating effect						Mediating type	Partial mediating effect				
Mediating effect/ Total effect	0.137						Mediating effect/ Total effect	0.206			Bootstrap test not significant -	

mediating effects on the path of “industrial policy→ innovation value”. Therefore, TD1 and TD2 are used as threshold variables to construct a panel threshold regression model [46] to further verify their possible threshold effect and intermediate link results.

Appendix B shows that in the threshold identification of TD1 and ID2, the subsidized policy has a greater bias toward short-term value realization of NEV enterprises (7a, 9a), while the non-subsidized policy is more strongly associated with long-term value (8b, 10b). All the variables exhibit a single threshold, and the threshold effect bias remains consistent with the overall impact results in Table 1 and the specific mediating effects in Table 2. These results verify the core link role of technology development behavior in the innovation value chain of NEV enterprises, but the threshold effects may vary along specific paths.

Table 3 presents the threshold regression results for each significant path. Regarding the influence process of the subsidized policy, radical innovation (TD1) exhibits a single threshold in the “Pol_{sub}→ SValue (11a)” and “Pol_{sub}→ LValue (11b)” paths, as does incremental innovation (TD2) in the “Pol_{sub}→ SValue (12a)” path, but all of these paths show marginal decreasing utility. Once the technology development level of NEV enterprises crosses these thresholds, the effect of the subsidized policy (Pol_{sub}) rapidly decreases from positive values. Even for the long-term value (11b), the TD1 threshold is lower, and the Pol_{sub} coefficient is no longer significant. The reason for this may be that in the early policy environment, which was dominated by direct monetary support, NEV enterprises had good expectations of “real money income” and “additional work, more pay.” Therefore, enterprises responded positively to the front-end R&D link (radical or incremental innovation) and realized innovation value, especially in short-term direct product output. However, as the industry develops to a certain scale, false innovation and low-level innovation inevitably increase. Some enterprises became less willing to innovate in a high-quality and substantial way, as is evident in NEV development in China. Coupled with the tightening of government subsidies, the marginal effect of the subsidized policy began to decrease, and the crowding-out effect on innovation value began to appear. Especially for the long-term market value that requires a longer realization time, the response to subsidy withdrawal is more sensitive.

In the influence process of the non-subsidized policy, there is a single threshold for TD1 and TD2 only in long-term value realization (LValue), but the threshold effect is the opposite. In the threshold effect of TD1, the non-subsidized policy effect (Pol_{non-sub}) is increasing, and the increment is significant (11c). In contrast, in TD2, the Pol_{non-sub} coefficient marginally decreased (12b). The reason for this may be that radical innovation is the primary source through which NEV enterprises enhance their long-term value. Under the influence of the non-

subsidized policy, which strengthens technology access and market competition, the reverse forcing effect on the high-level and radical innovation of NEV enterprises has been enhanced. This improvement in the quality and structure of previous technology development has been further significantly reflected in the latter value realization. However, incremental innovation itself may have a relatively small incentive effect on enterprises’ long-term innovation performance. An increase in incremental innovation inevitably squeezes out investment in radical innovation, coupled with the indirect monetary nature of the non-subsidized policy. Therefore, more incremental innovations may reduce the positive effect of the non-subsidized policy on the long-term value of NEV enterprises. Therefore, hypothesis H4 is confirmed.

4.3. A discussion

The empirical results presented above primarily illustrate the overall impact and dynamic changes of subsidized and non-subsidized policies on the innovation value chain of NEV enterprises, alongside the mediating role and threshold effect of the technology development behavior of NEV enterprises. The key conclusions drawn are linked to existing research findings. Firstly, concerning the impact of the subsidized policy, our findings suggest a positive influence on early-stage technological innovation and the expansion of the innovation market for NEVs, which aligns with established research perspectives [10,47]. Nonetheless, the gradual decrease in government subsidies since 2016 and the diminishing policy effects indicate that the subsidized policy for NEVs cannot be entirely phased out currently, necessitating essential adjustments [4,48]. Secondly, with regard to the non-subsidized policy, our study unveils that, through a synchronized comparison, the non-subsidized policy yields more stable innovation effects and excels in radical innovation and long-term value realization. This findings resonate with the notion that in the post-subsidy era, there is a requisite to reinforce the introduction and targeted implementation of non-subsidized policies for NEVs [27,49].

In these main conclusions, we have also uncovered novel and specific findings that complement existing literature. Based on the innovation value chain process of NEV enterprises, the subsidized policy is inclined towards generating short-term value (output value) by incentivizing various technology development behaviors, particularly incremental innovation, within NEV enterprises. Nonetheless, the short-term incentive effect of the subsidized policy exhibits a specific threshold range, whereby the subsidy amount must be set at an optimal level to achieve the best solution for the two stages of the innovation value chain. Conversely, the non-subsidized policy has the potential to stimulate more radical innovation, with the effectiveness of the “non-subsidized policy→ long-term value (market value)” pathway increasing in correlation with the presence of radical innovations.

Table 3
Threshold regression results (TD1, TD2).

Threshold types	TD1			TD2	
	(11a) Pol _{sub} → SValue	(11b) Pol _{sub} → LValue	(11c) Pol _{non-sub} → LValue	(12a) Pol _{sub} → SValue	(12b) Pol _{non-sub} → LValue
Pol _{sub} (TD1≤4.875)	1.225*** (0.282)				
Pol _{sub} (TD1>4.875)	0.205** (0.098)				
Pol _{sub} (TD1≤2.639)		0.653*** (0.197)			
Pol _{sub} (TD1>2.639)		0.035 (0.059)			
Pol _{non-sub} (TD1≤3.045)			0.023 (0.038)		
Pol _{non-sub} (TD1>3.045)			0.225*** (0.052)		
Pol _{sub} (TD2≤4.970)				1.391*** (0.298)	
Pol _{sub} (TD2>4.970)				0.238** (0.097)	
Pol _{non-sub} (TD2≤4.234)					0.270*** (0.057)
Pol _{non-sub} (TD2>4.234)					0.138** (0.054)
Control variables	Yes				
Constant	5.146*** (0.517)	2.950*** (0.318)	2.487*** (0.292)	5.162*** (0.510)	2.511*** (0.288)
Firm FE	Yes				
Adj. R ²	0.262	0.166	0.167	0.273	0.184

Moreover, we have identified more prominent issues that exist. On the one hand, within the innovation value chain framework, the overall mediating impact of technology development behavior on NEV enterprises is relatively limited, resulting in inadequate innovation transfer efficiency. The incentivizing effects of policies on front-end technology development have not been fully leveraged to translate into innovation value at the back-end. On the other hand, the mediating effects of diverse technology development behaviors of enterprises reveal that radical innovation and incremental innovation each possess distinct emphases, with a stronger correlation to short-term innovation value realization and a narrower influence on long-term value. This pattern is closely tied to the observation that the innovation of China's NEV enterprises in the early stages leans more towards incremental innovation, neglecting the pursuit of high-quality development. Consequently, the outcomes highlight the prevailing challenge in China's NEV industry of being large but not strong, where enterprises exhibit surplus short-term output but encounter obstacles in achieving long-term value.

4.4. Heterogeneity study

Previous research has highlighted that under the influence of industrial policies, factors such as internal ownership of enterprises and external environments may exert diverse impacts on enterprises' innovation activities [20,50]. Hence, we also incorporate these two factors to investigate the potential heterogeneous effects of subsidized and non-subsidized policies.

(1) Influence of internal ownership. We divided the research sample into two groups: state-owned and non-state-owned vehicle enterprises. The results presented in Table 4 indicate that subsidized and non-subsidized policies have different impacts on front-end technology development behaviors. The subsidized policy (Pol_{sub}) performed better among state-owned vehicle enterprises, particularly in stimulating incremental innovation (TD2, 13b). The non-subsidized policy ($Pol_{non-sub}$) is more effective among non-state-owned vehicle enterprises and provides the greatest incentives for radical innovation (TD1, 15a). With respect to back-end innovation value realization, influenced by front-end outcomes, the subsidized policy has the strongest incentive effect on the short-term product output of state-owned vehicle enterprises (SValue, 14a), while the non-subsidized policy has a more balanced effect on the value realization of non-state-owned vehicle enterprises, particularly in the long-term value (LValue, 16b). These results further highlight the significance of adjusting the NEV policy in a non-subsidy direction, and the design and implementation of future policy tools should also take into account the differences in the policy responses of state-owned and non-state-owned vehicle enterprises. Moreover, to better examine the coefficient differences between the groups, Chow tests were conducted when TD1, TD2, SValue, and LValue indicators were used as the dependent variables, respectively. The test results showed that the p-values were 0.011, 0.032, 0.003, and 0.004, which were all significant at the 5 % or 1 % level. This demonstrates the presence of actual coefficient differences among groups.

(2) Influence of the external environment. We selected the innovation atmosphere index of the region where the headquarters of NEV enterprises are located as the distinguishing variable.⁵ The results presented in Table 5 indicate that in the high-regional innovation atmosphere group, the non-subsidized policy ($Pol_{non-sub}$) has a significantly better effect than the subsidized policy (Pol_{sub}) at the

front-end of the innovation value chain (17a, 17b). At the back-end, the effects of the two policies have different priorities, as reflected in the paths of Pol_{sub} -SValue (18a) and $Pol_{non-sub}$ -LValue (18b). Therefore, for NEV enterprises operating in areas with a high innovation atmosphere, the vigorous implementation of non-subsidized policies and the deepening of policy effects to the back-end of the innovation value chain will yield positive policy outcomes. In contrast, in the low-regional innovation atmosphere group, the difference between the two policies is more pronounced. The subsidized policy (Pol_{sub}) has an overall better effect than the non-subsidized policy, especially in terms of front-end technology development (19a, 19b) and short-term value outcomes (20a). Therefore, for NEV enterprises in areas with a low innovation atmosphere, whether the subsidized policy should be withdrawn completely requires further discussion. Overall, these results show that it is equally important to implement the incentive effect of policies according to local conditions while dynamically adjusting the two policies. Similarly, when TD1, TD2, SValue, and LValue indicators were used as dependent variables, the results of the Chow tests showed p-values of 0.000, 0.006, 0.001, and 0.004, indicating that actual coefficient differences among groups also exist.

4.5. Robustness tests

We conducted some necessary robustness tests. First, we replaced the core variables. For the technology development indicator, we replaced the original patent application indicator with the lagged one-period patent authorization amount (invention and non-invention patent authorization, ATD1 and ATD2) [51]. For the short-term innovation value indicator, we replaced it with the enterprise's main business income (ASValue) [1]; for the long-term innovation value indicator, we replaced it with the enterprise competitiveness index $(= (\text{total profit} + \text{total tax} + \text{interest expense}) / \text{average total assets})$ (ALValue) [52]. The results in Appendix C (21a)-(22b) show that the effect trends of the two policy variables remain consistent with the benchmark regression results.

Next, we changed the sample time interval. On the one hand, we took the year 2010 as the starting point for the entry of NEVs into China's seven strategic emerging industries, expanding the sample interval to 2010–2022. On the other hand, considering the impact of COVID-19 on the NEV industry, we excluded data from 2020 to 2022, narrowing the sample interval to 2012–2019. The results in Appendix C (23a)-(26b) show that the overall effects and comparative results of the two policy variables remain robust.

Third, we controlled for macro-environmental factors. In addition to controlling for existing internal enterprise indicators, we conducted a new regression by including other external control factors, such as the regional innovation environment (regional innovation atmosphere, footnote 5) and the market development level (annual total sales volume of NEVs, 10,000 units). The results in Appendix C (27a)-(28b) show that the signs and significance of the two policy variables did not change significantly, indicating that the results of this study remain robust.

5. Conclusions and policy implications

5.1. Main conclusions

1) Both subsidized and non-subsidized policies exhibit an overall positive impact on technological innovation within NEV enterprises, as seen through the innovation value chain. Specifically, in the previous stage of enterprise technology development, the subsidized policy demonstrates a relatively stronger but volatile effect, and is more likely to trigger incremental innovation. The non-subsidized policy has a relatively lower but stable effect, and continues to provide strong incentives for radical innovation. In the latter stage of innovation value realization, both policies have a more pronounced effect

⁵ To be specific, we have used the comprehensive index of the regional innovation capacity of the provinces where the headquarters of NEV enterprises are located in the *China Regional Innovation Capacity Evaluation Report* as a measure to divide the research samples into two groups: high-regional innovation atmosphere and low-regional innovation atmosphere.

Table 4

Differences in innovation result between state-owned and non-state-owned enterprises.

Variables	State-owned enterprises group				Non-state-owned enterprises group			
	(13a) TD1	(13b) TD2	(14a) SValue	(14b) LValue	(15a) TD1	(15b) TD2	(16a) SValue	(16b) LValue
Pol _{sub}	0.263 (0.183)	0.347** (0.171)	0.123* (0.072)	0.022 (0.049)	0.085 (0.177)	0.088** (0.041)	0.071* (0.046)	0.137* (0.083)
Pol _{non-sub}	−0.068 (0.155)	−0.173 (0.144)	0.027 (0.061)	0.161 (0.042)	0.310*** (0.136)	0.108** (0.048)	0.068* (0.041)	0.278*** (0.104)
Control variables	Yes							
Constant	3.000** (1.382)	3.612*** (1.287)	6.448*** (0.544)	2.627*** (0.371)	11.459** (4.836)	11.815** (5.161)	−3.649 (3.208)	−0.464 (1.363)
Firm FE	Yes							
N	154	154	154	154	106	106	106	106
Adj. R ²	0.844	0.843	0.829	0.680	0.859	0.777	0.713	0.583

Table 5

Differences in innovation result between high- and low-regional innovation atmosphere.

Variables	High-regional innovation atmosphere group				Low-regional innovation atmosphere group			
	(17a) TD1	(17b) TD2	(18a) SValue	(18b) LValue	(19a) TD1	(19b) TD2	(20a) SValue	(20b) LValue
Pol _{sub}	−0.051 (0.201)	0.050 (0.242)	0.428*** (0.130)	0.118* (0.069)	0.693** (0.343)	0.747** (0.296)	0.030* (0.017)	0.008 (0.087)
Pol _{non-sub}	0.529*** (0.172)	0.521** (0.207)	0.218* (0.116)	0.243*** (0.059)	−0.333** (0.156)	−0.260* (0.135)	−0.023 (0.077)	0.104* (0.060)
Control variables	Yes							
Constant	−2.139 (1.541)	−2.226 (1.854)	2.063** (0.998)	2.516*** (0.530)	−0.884 (11.121)	24.7471*** (9.594)	−4.080 (5.494)	−0.459 (2.812)
Firm FE	Yes							
N	128	128	128	128	132	132	132	132
Adj. R ²	0.905	0.837	0.884	0.649	0.812	0.796	0.800	0.557

on the short-term innovation value (output value) of NEV enterprises, but a lower or even negative effect on the long-term value (market value).

- The mediation effect and threshold effect tests reveal that within the overall innovation value chain framework, the technology development behavior of NEV enterprises plays a significant mediating role, but the specific mediation pathways and threshold effects are not consistent. Notably, the overall mediation effect is relatively low, indicating insufficient innovation transmission efficiency, as the incentive effects of the two policies on front-end technology development have not fully translated to back-end innovation value.
- The heterogeneity tests indicate that, concerning enterprise ownership grouping, the innovation incentive effect of the subsidized policy is more pronounced in state-owned vehicle enterprises, particularly in the context of incremental innovation behavior and short-term value realization. The non-subsidized policy demonstrates greater effectiveness in non-state-owned vehicle enterprises, encouraging radical innovation and long-term value realization. Regarding the external regional innovation atmosphere grouping, the subsidized policy remains a crucial support for vehicle enterprises operating in regions with a low innovation atmosphere. On the other hand, the non-subsidized policy is likely to have a more positive impact on vehicle enterprises operating in regions with a high innovation atmosphere.

5.2. Policy implications

- The NEV subsidized policy cannot be completely withdrawn at present, but adjustments are necessary. First, we need to convert direct consumption subsidies and universal subsidy forms, and use market-oriented methods to screen out capable and willing vehicle enterprises, providing them with special innovation grants. Simultaneously, we must establish the threshold value and reasonable scale of innovation subsidies to align policy incentive objectives with enterprises' innovation demands. Furthermore, there is a need to

improve the effectiveness of innovation subsidies by ensuring an optimal match between government subsidy tools and the enterprise innovation value chain process.

- The NEV non-subsidized policy should be the primary focus of policy implementation in the post-subsidy era. First, enhancing the effectiveness of existing non-subsidized policies is important for promoting high-level innovation. For instance, we can maintain strict entry thresholds and dynamic guidance for the government's recommended model catalogs to encourage NEV enterprises to innovate proactively. Meanwhile, improving accounting standards and the dual-credit ratio for NEVs, as well as enforcing NEV priority procurement and mandatory procurement standards, can enhance the government's ability to guarantee innovation by vehicle enterprises. Furthermore, considering the unique characteristics of the innovation value chain, we can explore combinations of policy tools across different categories, including both subsidized and non-subsidized policies. This can maximize the effectiveness of policy mixes and address the shortcomings of existing policies, which primarily focus on providing short-term output incentives but may lack sufficient incentives for long-term value creation.
- The adjustment of policies in the NEV industry and the optimization of the innovation value chain should also consider the heterogeneous outcomes resulting from the internal characteristics and external environment of enterprises. On the one hand, the government should focus on guiding the innovation behavior of state-owned vehicle enterprises by reducing their overreliance on policy resources, such as government subsidies. For non-state-owned vehicle enterprises, it is essential to nurture their drive for high-level innovation and market competitiveness through the implementation of non-subsidized policies, such as the effective enforcement of the dual-credit competition system and the establishment of a fair policy support environment. On the other hand, when considering vehicle enterprises operating in diverse regional environments, the government should adopt a differentiated approach to innovation-driven strategies. For vehicle enterprises in regions with a high innovation

atmosphere, the government should enhance and optimize the non-subsidized policy framework, incorporating key measures like market access and the dual-credit system. It is essential to ensure the appropriateness of technical indicators and standards, motivating enterprises to pursue continuous innovation and generate long-term value. For vehicle enterprises in regions with a low innovation atmosphere, the government should prioritize enhancing the innovation environment and bolstering support for innovation foundations. This can be accomplished through targeted innovation subsidies and public procurement. By taking these steps, the government can effectively stimulate enterprises' enthusiasm for innovation and drive the formation of the overall industry value chain.

This study also has several limitations. First, the theoretical foundation and measurement of policy effectiveness for two types of NEV policies, subsidized and non-subsidized, require further refinement. While the division between subsidized and non-subsidized policies holds theoretical exploratory significance as a policy system for an emerging industry, this study is inevitably constrained by certain data and sample limitations. In the future, the research perspective could be broadened to encompass the entire industry chain of enterprises, while integrating other research methods, such as typical case studies, to enhance the comparison of the innovation effects of the two policies. Second, this study primarily selected indicators such as patent applications, new product output, and the market value of NEV enterprises to measure the results at different stages of the innovation value chain. In the future, more refined indicators could be considered to more accurately measure the technology development behaviors and innovation value levels of enterprises, and to further explore the relationship between NEV industry policies and enterprise innovation process. Furthermore, the combined implementation of different policy tools in the two policies, as

well as the verification of their effects in other emerging industries or regional markets, can also be important directions for future research.

Disclosure statement

No potential conflict of interest was reported by the authors.

CRediT authorship contribution statement

Shufeng Qin: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yongqing Xiong:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Variable descriptive statistics and correlation coefficients

Variables	Mean	SD	Min	Max	1	2	3	4	5	6	7	8	9	10	11	12
1. SValue	4.774	1.279	1.386	7.268	1											
2. LValue	1.280	0.331	0.684	2.471	−0.175***	1										
3. Pol _{sub}	9.544	19.932	0.036	114.391	0.199***	−0.125*	1									
4. Pol _{non-sub}	2.231	0.688	0	3	0.304***	0.093	0.217***	1								
5. TD1	4.553	1.935	0	8.203	0.478***	0.082	0.329***	0.441***	1							
6. TD2	5.274	1.669	0	7.803	0.409***	−0.078	0.271***	0.290***	0.882***	1						
7. Size	5.729	1.338	2.460	9.200	0.293***	−0.179**	0.587***	0.434***	0.628***	0.580***	1					
8. Age	2.935	0.334	0.811	3.388	−0.007	−0.052	0.002	0.217***	−0.116*	−0.302***	−0.160***	1				
9. Own	0.601	0.491	0	1	0.546***	−0.070	0.115*	0.237***	0.170***	0.123*	0.046	−0.120*	1			
10. ROA	0.023	0.066	−0.548	0.157	0.182**	0.114*	0.238***	−0.072	0.251***	0.289***	0.349***	−0.258***	0.015	1		
11. ALR	0.620	0.145	0.291	0.977	0.202***	0.144**	−0.058	0.061	−0.077	−0.162**	−0.242***	0.012	0.417***	−0.283***	1	
12. HC	0.197	0.075	0.024	0.596	0.027	0.018	−0.244***	0.167***	0.190***	0.192***	0.074	0.071	−0.013	−0.047	−0.263***	1

Appendix B

Threshold effect identification (TD1, TD2)

Dependent variables	Independent variables	Threshold model	Threshold estimate	F-value	p-value	10 %	5 %	1 %
Threshold variable: TD1								
Pol _{sub}	(7a) SValue	Single threshold	4.875	15.03**	0.047	11.865	14.528	20.001
		Double threshold	6.931	4.86	0.560	10.283	12.552	17.581
	(7b) LValue	Single threshold	2.639	12.62*	0.063	11.242	13.612	18.618
		Double threshold	5.724	7.68	0.287	10.662	12.033	18.985
Pol _{non-sub}	(8a) SValue	Single threshold	6.617	10.16	0.140	11.736	13.642	18.575
		Single threshold	3.045	12.82**	0.030	9.078	11.005	15.053
	(8b) LValue	Double threshold	7.245	6.46	0.287	8.700	10.610	14.065

(continued on next page)

(continued)

Dependent variables	Independent variables	Threshold model	Threshold estimate	F-value	p-value	10 %	5 %	1 %
Threshold variable: TD2								
Pol _{sub}	(9a) SValue	Single threshold	4.970	18.04**	0.017	12.372	15.413	18.946
		Double threshold	6.360	6.98	0.393	10.903	13.696	17.454
	(9b) LValue	Single threshold	3.871	9.23	0.130	10.030	12.403	16.105
		Single threshold	6.629	7.23	0.383	13.776	16.753	27.122
	(10a) SValue	Single threshold	4.234	17.46**	0.017	10.961	12.599	18.905
		Double threshold	7.374	6.97	0.257	10.033	12.939	16.809
Pol _{non-sub}	(10b) LValue							

Appendix C

Robustness test results

Variables	Core variables replacement				Sample interval replacement (2010–2022)			
	(21a) ATD1	(21b) ATD2	(22a) ASValue	(22b) ALValue	(23a) TD1	(23b) TD2	(24a) SValue	(24b) LValue
Pol _{sub}	0.048 (0.071)	0.134** (0.076)	0.105** (0.040)	−0.004 (0.005)	0.087 (0.170)	0.204* (0.109)	0.370*** (0.122)	−0.036 (0.060)
Pol _{non-sub}	0.426** (0.166)	0.219 (0.184)	0.093 (0.094)	0.031** (0.014)	0.162* (0.089)	0.102 (0.091)	0.094 (0.069)	0.084*** (0.025)
Control variables	Yes				Yes			
Constant	−2.212** (0.977)	3.888*** (1.086)	−0.724 (0.553)	0.029 (0.028)	−0.469 (1.175)	4.389*** (1.336)	3.535*** (1.0723)	2.832*** (0.348)
Firm FE	Yes				Yes			
N	260	260	260	260	304	304	304	304
Adj. R ²	0.860	0.771	0.820	0.468	0.858	0.763	0.681	0.356
Variables	Sample interval replacement (2012–2019)				Macro-environmental factors control			
	(25a) TD1	(25b) TD2	(26a) SValue	(26b) LValue	(27a) TD1	(27b) TD2	(28a) SValue	(28b) LValue
Pol _{sub}	0.098 (0.163)	0.375** (0.199)	0.288*** (0.122)	0.019 (0.067)	0.108 (0.171)	0.293** (0.141)	0.284** (0.109)	0.033 (0.061)
Pol _{non-sub}	0.312** (0.153)	0.152* (0.086)	0.073 (0.048)	0.056*** (0.027)	0.215** (0.085)	0.137 (0.091)	0.050 (0.048)	0.085*** (0.028)
Control variables	Yes				Yes			
Constant	0.970 (1.037)	4.492*** (1.354)	3.773*** (0.671)	2.847*** (0.268)	0.883 (1.691)	4.827*** (1.818)	1.048 (1.003)	2.231*** (0.571)
Firm FE	Yes				Yes			
N	192	192	192	192	260	260	260	260
Adj. R ²	0.909	0.816	0.807	0.415	0.840	0.750	0.864	0.348

References

[1] Zhang T, Li S, Li Y, Wang W. Evaluation of technology innovation efficiency for the listed NEV enterprises in China. *Econ Anal Pol* 2023;80:1445–58.

[2] Liu Q, Wen X, Peng H, Cao Q. Key technology breakthrough in new energy vehicles: Configuration path evolution from innovative ecosystem perspective. *J Clean Prod* 2023;423:138635.

[3] Jiang Z, Xu C. Policy incentives, government subsidies, and technological innovation in new energy vehicle enterprises: evidence from China. *Energy Pol* 2023;177:113527.

[4] Du Y, Guo Z, Bao H. Smooth sailing ahead? Policy options for China's new energy vehicle industry in the post-subsidy era. *Energy Res Social Sci* 2024;107:103359.

[5] Ye RK, Gao ZF, Fang K, Liu KL, Chen JW. Moving from subsidy stimulation to endogenous development: a system dynamics analysis of China's NEVs in the post-subsidy era. *Technol Forecast Soc Change* 2021;168:120757.

[6] Qin S, Xiong Y. Innovation strategies of Chinese new energy vehicle enterprises under the influence of non-financial policies: effects, mechanisms and implications. *Energy Pol* 2022;164:112946.

[7] Zhang L, Lin Y, Zhou S, Qin Q. Evolution and trends of new energy vehicle policy research hotspots: an analysis based on a Citespace knowledge graph. *Environ Sci Pollut Control Ser* 2024;31(5):7728–50.

[8] Liu Q, Wen X, Cao Q. Multi-objective development path evolution of new energy vehicle policy driven by big data: from the perspective of economic-ecological-social. *Appl Energy* 2023;341:121065.

[9] Sun X, Liu X, Wang Y, Yuan F. The effects of public subsidies on emerging industry: an agent-based model of the electric vehicle industry. *Technol Forecast Soc Change* 2019;140:281–95.

[10] Wu Y, Li X, Zhang C, Wang S. The impact of government subsidies on technological innovation of new energy vehicle enterprises: from the perspective of industry chain. *Environ Dev Sustain* 2023;1–19.

[11] Qin S, Xiong Y. Differential impact of subsidised and nonsubsidised policies on the innovation of new energy vehicle enterprises: evidence from China. *Asian J Technol Innovat* 2023;31(2):260–83.

[12] Liu D, Chen T, Liu X, Yu Y. Do more subsidies promote greater innovation? Evidence from the Chinese electronic manufacturing industry. *Econ Modell* 2019; 80:441–52.

[13] Xu XL, Chen HH. Exploring the innovation efficiency of new energy vehicle enterprises in China. *Clean Technol Environ Policy* 2020;22(8):1671–85.

[14] Li Q, Wang M, Xiangli L. Do government subsidies promote new-energy firms' innovation? Evidence from dynamic and threshold models. *J Clean Prod* 2021;286: 124992.

[15] Chen X, Liu Z, Zhu Q. Reprint of "Performance evaluation of China's high-tech innovation process: analysis based on the innovation value chain". *Technovation* 2020;94:102094.

[16] Roper S, Du J, Love JH. Modelling the innovation value chain. *Res Pol* 2008;37 (6–7):961–77.

[17] Dong F, Liu Y. Policy evolution and effect evaluation of new-energy vehicle industry in China. *Resour Pol* 2020;67:101655.

[18] Yang T, Xing C, Li X. Evaluation and analysis of new-energy vehicle industry policies in the context of technical innovation in China. *J Clean Prod* 2021;281: 125126.

[19] Liu C, Liu Y, Zhang D, Xie C. The capital market responses to new energy vehicle (NEV) subsidies: an event study on China. *Energy Econ* 2022;105:105677.

[20] Wang Z, Li X, Xue X, Liu Y. More government subsidies, more green innovation? The evidence from Chinese new energy vehicle enterprises. *Renew Energy* 2022; 197:11–21.

- [21] Wang N, Tang L, Zhang W, Guo J. How to face the challenges caused by the abolishment of subsidies for electric vehicles in China? *Energy* 2019;166:359–72.
- [22] Shao W, Yang K, Bai X. Impact of financial subsidies on the R&D intensity of new energy vehicles: a case study of 88 listed enterprises in China. *Energy Strategy Rev* 2021;33:100580.
- [23] Chen Y, Ni L, Liu K. Does China's new energy vehicle industry innovate efficiently? A three-stage dynamic network slacks-based measure approach. *Technol Forecast Soc Change* 2021;173:121161.
- [24] Hardman S. Understanding the impact of reoccurring and non-financial incentives on plug-in electric vehicle adoption - a review. *Transport Res Pol Pract* 2019;119: 1–14.
- [25] Li X, Liu Y, Jing B. Research on the impact of industrial support policies on the promotion of new energy vehicles in China. *Manag Rev* 2022;34(3):55–65.
- [26] Qadir SA, Ahmad F, Al-Wahedi AMA, Iqbal A, Ali A. Navigating the complex realities of electric vehicle adoption: a comprehensive study of government strategies, policies, and incentives. *Energy Strategy Rev* 2024;53:101379.
- [27] Wang X, Xiong Z, Li X, Xiong Y. How do nonsubsidized incentive affect enterprises' innovation choices? A case from the new energy vehicle industry in China. *Technol Forecast Soc Change* 2024;204:123445.
- [28] Ma M, Meng W, Huang B, Li Y. The influence of dual credit policy on new energy vehicle technology innovation under demand forecast information asymmetry. *Energy* 2023;271:127106.
- [29] He H, Li S, Wang S, Zhang C, Ma F. Value of dual-credit policy: evidence from green technology innovation efficiency. *Transport Pol* 2023;139:182–98.
- [30] Ou S, Lin Z, He X, Przesmitzki S, Bouchard J. Modeling charging infrastructure impact on the electric vehicle market in China. *Transport Res Transport Environ* 2020;81:102248.
- [31] Ledna C, Muratori M, Brooker A, Wood E, Greene D. How to support EV adoption: tradeoffs between charging infrastructure investments and vehicle subsidies in California. *Energy Pol* 2022;165:112931.
- [32] Wang X, Jiang C. The signal effect of new energy vehicles promotion on enterprise innovation. *Complexity* 2021;2021:9920740.
- [33] Yao X, Ma S, Bai Y, Jia N. When are new energy vehicle incentives effective? Empirical evidence from 88 pilot cities in China. *Transport Res Pol Pract* 2022;165: 207–24.
- [34] Bleda M, Chicot J. The role of public procurement in the formation of markets for innovation. *J Bus Res* 2020;107:186–96.
- [35] Xiong Y, Qin S. Differences in consumers' product attribute preferences and willingness to pay for new energy vehicles: a comparison between the license plate- and non-license plate-restricted cities in China. *International Journal of Sustainable Transportation* 2022;16(1):57–72.
- [36] Zhao X, Li X, Wu Y, Qiao L, Zhang C. Assessment of the effects of China's new energy vehicle industry policies: from the perspective of moderating effect of consumer characteristics. *Environ Dev Sustain* 2023:1–22.
- [37] Lin S, Xiao L, Wang X. Does air pollution hinder technological innovation in China? A perspective of innovation value chain. *J Clean Prod* 2021;278:123326.
- [38] Yuan X, Li X. The evolution of the industrial value chain in China's high-speed rail driven by innovation policies: a patent analysis. *Technol Forecast Soc Change* 2021;172:121054.
- [39] Song Y, Hu J, Qi Y. An empirical study of establishment of the time-lag effect model of innovation value transformation. *Sci Res Manag* 2022;43(3):192–200.
- [40] Hao X, Chen F, Chen Z. Does green innovation increase enterprise value? *Bus Strat Environ* 2022;31(3):1232–47.
- [41] Bronzini R, Piselli P. The impact of R&D subsidies on firm innovation. *Res Pol* 2016;45(2):442–57.
- [42] Hou Q, Chen Z, Teng M. Today's baton and tomorrow's vision: the effect of strengthening patent examination system on corporate innovation strategies. *J Bus Res* 2022;144:614–26.
- [43] Han J, Jiang C, Liu R. Does intelligent transformation trigger technology innovation in China's NEV enterprises? *Energy* 2023;270:126823.
- [44] Zhao Y, Jian Z, Du Y. How can China's subsidy promote the transition to electric vehicles? *Renew Sustain Energy Rev* 2024;189:114010.
- [45] Xiong Y, Qin S. Differences in the effects of China's new energy vehicle industry policies on market growth from the perspective of policy mix. *Energy Environ* 2021;32(3):542–61.
- [46] Hansen BE. Threshold effects in non-dynamic panels: estimation, testing, and inference. *J Econom* 1999;93(2):345–68.
- [47] Ren H, Liu Z. The impact of government subsidies on R&D investment of listed companies in China's new energy vehicle industry-based on the perspective of industrial chain. *Asian J Technol Innovat* 2023:1–27.
- [48] Wang Y, Liao Z. Functional industrial policy mechanism under natural resource conflict: a case study on the Chinese new energy vehicle industry. *Resour Pol* 2023; 81:103417.
- [49] Kong D, Xia Q, Xue Y, Zhao X. Effects of multi policies on electric vehicle diffusion under subsidy policy abolishment in China: a multi-actor perspective. *Appl Energy* 2020;266:114887.
- [50] Wang X, Zou H, Zheng Y, Jiang Z. How will different types of industry policies and their mixes affect the innovation performance of wind power enterprises? Based on dual perspectives of regional innovation environment and enterprise ownership. *J Environ Manag* 2019;251:109586.
- [51] Shen H, Hou F. Trade policy uncertainty and corporate innovation evidence from Chinese listed firms in new energy vehicle industry. *Energy Econ* 2021;97:105217.
- [52] Li G, Wang X, Su S, Su Y. How green technological innovation ability influences enterprise competitiveness. *Technol Soc* 2019;59:101136.

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